

Privacy, Environment, Accountability & Workforce Ethics

Privacy & Surveillance

AI-powered surveillance systems — facial recognition in public spaces, predictive policing, social media monitoring — raise profound privacy concerns. The capability to identify, track, and profile individuals at scale without their knowledge or consent challenges fundamental rights.

- Mass facial recognition: China's Social Credit System; police use of Clearview AI in democracies.
- Workplace surveillance: AI monitoring keystrokes, eye-tracking, communication sentiment.
- Ad targeting: detailed behavioural profiles built from data aggregated across devices and platforms.

Key tension: surveillance AI can enhance public safety AND enable authoritarian control — context and oversight are everything.

Environmental Impact

Training large AI models consumes enormous energy and water. This environmental cost is increasingly visible as AI scales up:

- GPT-3 training emitted ~552 tonnes of CO₂ — equivalent to ~120 petrol cars driven for a year.
- Inference (running) a large model is less energy-intensive per query but happens billions of times daily.
- Data centres require enormous water for cooling — some models consume millions of litres per day.
- Mitigation: model compression (smaller, efficient models), renewable energy data centres, mixture-of-experts architectures.

Accountability — The Unresolved Questions

Who is responsible?: When an AI makes a harmful decision — the developer, the deployer, the user, or the AI itself?

Explainability gap: Many high-stakes AI decisions (credit, medical) cannot be explained in human-understandable terms.

Regulatory lag: AI capabilities advance faster than governments can legislate — the EU AI Act is a landmark attempt.

Workforce Ethics

AI automation displaces some roles while creating others. The ethical question is not just economic but social — how do we ensure the transition is equitable?

- Routine cognitive tasks (data entry, basic analysis, customer service) face the highest automation risk.
- Roles requiring creativity, empathy, complex judgment, and physical dexterity are more resilient.
- Reskilling is the policy response — lifelong learning, portable benefits, wage support during transitions.

Estimated: AI will automate ~15% of global work tasks by 2030 (McKinsey) while creating new roles in AI oversight, data curation, and human-AI collaboration.

Key Takeaways

- Privacy: AI surveillance requires strong legal frameworks and democratic oversight.
- Environment: large model training has significant carbon and water footprints.
- Accountability: liability frameworks and explainability requirements are still evolving.
- Workforce: reskilling and equitable transition policies are essential.

Quick-Revision Questions

- What is the 'explainability gap' and why does it matter for accountability?
- Describe two environmental costs of large AI model training.
- Which types of jobs are most and least at risk from AI automation?